R&D and Innovation System: Knowledge/Technology Exchange Issues

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Turkish Ecosystem…

Turkey, has been a pioneer in the extended version of the definition of "Tech Transfer" as "Knowledge Transfer".

- Knowledge Transfer / Exchange was an integral part of The Secintific and Technological Research Council of Turkey (TUBITAK) TTO Program
- The extended scope is now included in the comprehensive definition of the Profession….

University Industry Collaboration Centers Platform (USIMP) has been working very closely with the National Professional Proficiency Institute

- to define the technology transfer profession and
- to identify the core competencies for the technology transfer expert (level 6), in the last three years.

The official definition has been published in the Official Gazette on June 6th, 2018.

Today, the profession has also significantly expanded…

- TTOs (*interface organisations*) now include government labs and hospitals and their industrial counterparts.
- They are responsible for ***business development and startup formation, industry collaborations and partnerships***.
- Some have a sharp focus on economic development and,

Thus, the formal training for K/TT Professionals now includes, the scope that matches the needs of the sector.

K/TT professionals are no longer focused …

solely on the evaluation, protection and commercialization of academic research.

But…

- … are also running incubators, forming start-ups and in industry
  developing medicines and smart technologies.
- … are entrepreneurs, marketers, venture capitalists, relationship managers and legal experts.
- … are all ***part of the process that links the development of innovations from research to the public where they provide societal benefit.***

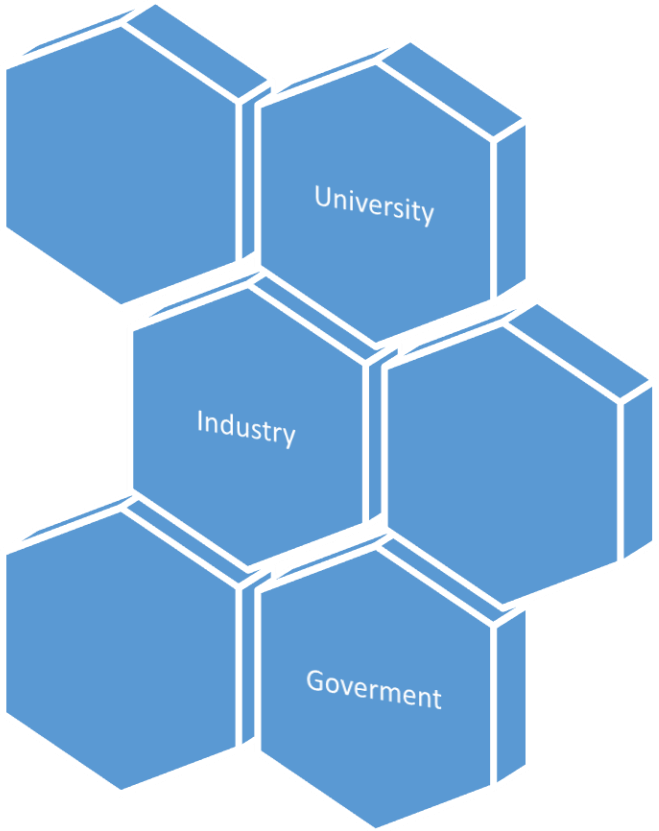
Types of COLLABORATIONs with Industry

1. Training (Academician is trainer)
2. Technical Reports
3. Advisor
4. Field Study
5. Authority
6. Joint Projects

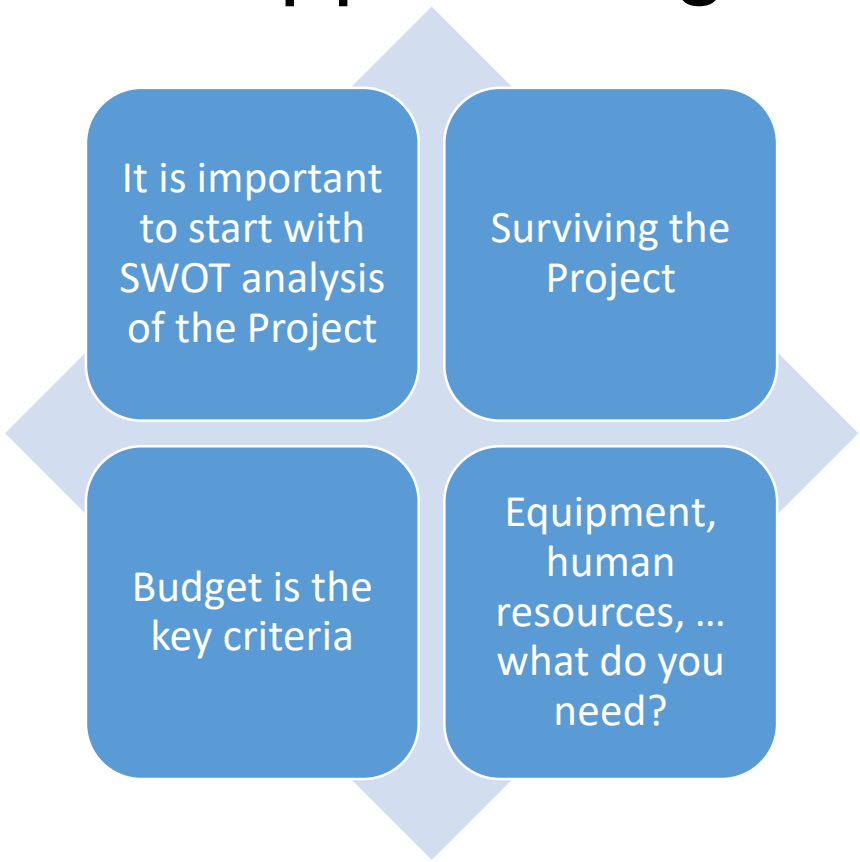
Types of Joint Projects

- Research based
  - Mostly University is the leader in the project
- Innovation based
  - Mostly Industry is the leader in the project
- Consultancy based
  - Industry is the responsible body and university is the driver

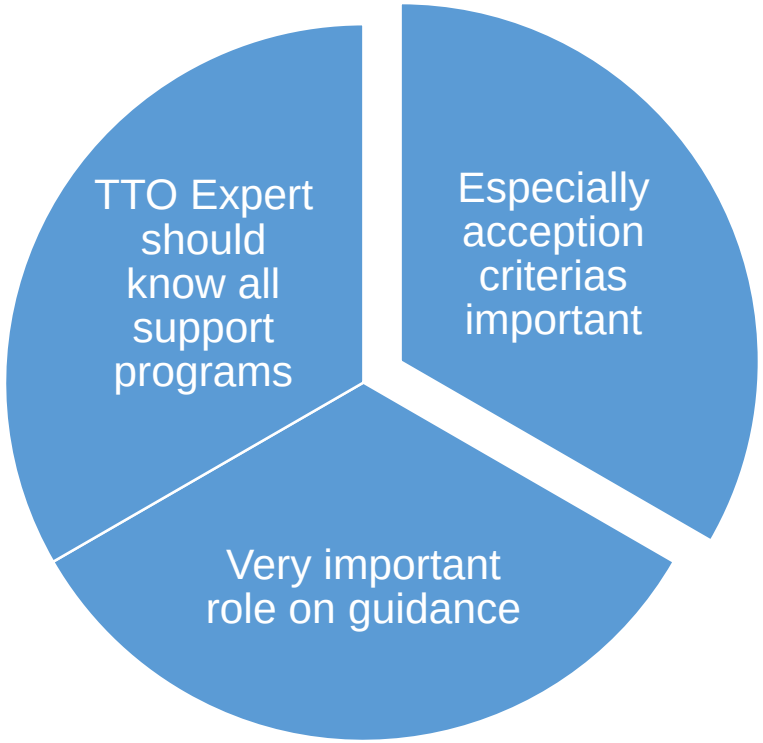
Joint Projects Support Programs



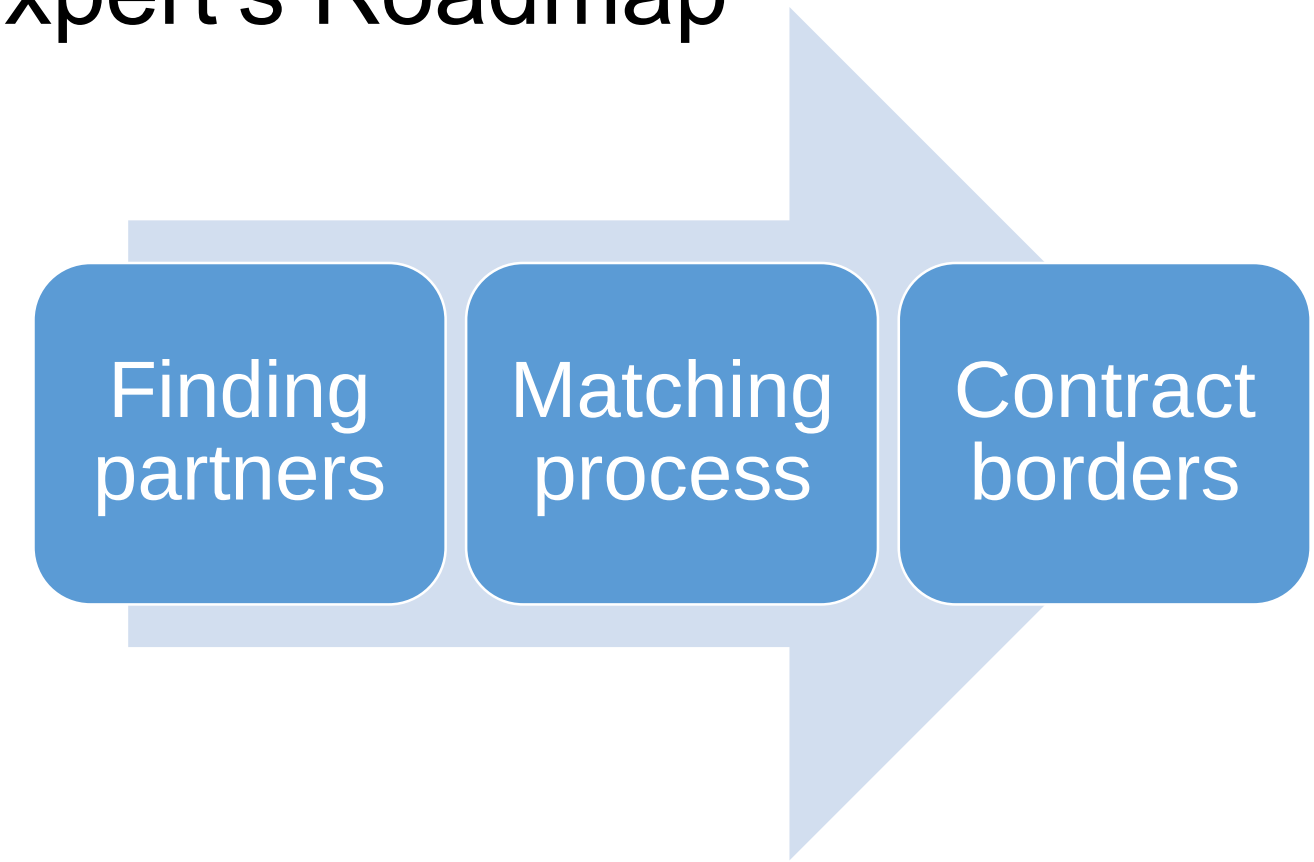
Selecting Best Support Program To Apply



Role of TTO Expert



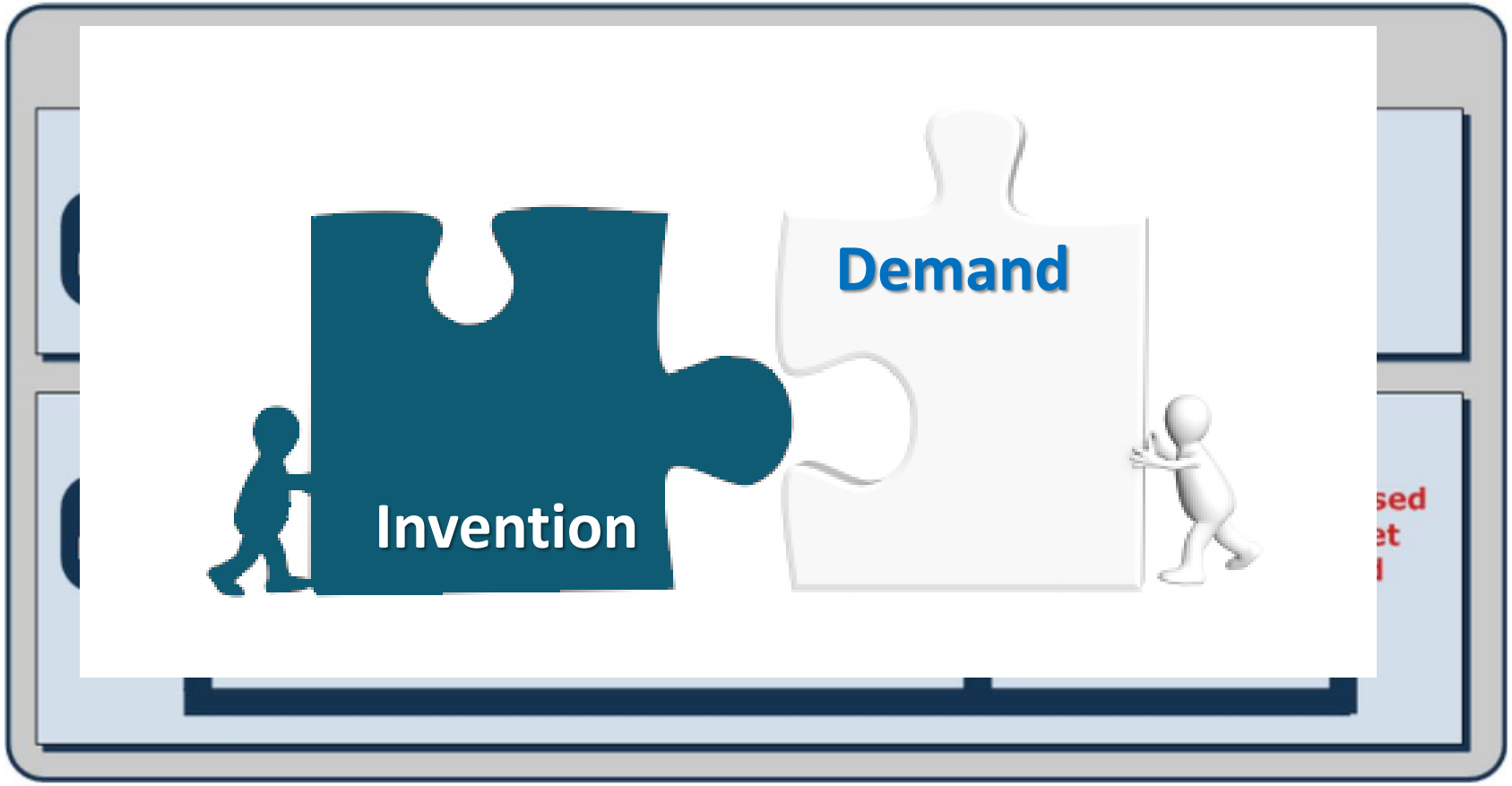
Developing Contract Based Researches
TTO Expert's Roadmap



How to Trigger Collaboration



Industrial Engagement: One OR Both?



Martin, Michael J.C. (1994). Managing Innovation and Entrepreneurship in Technology-based Firms. Wiley-IEEE.

Collaboration with Industry

Industry brings funding and

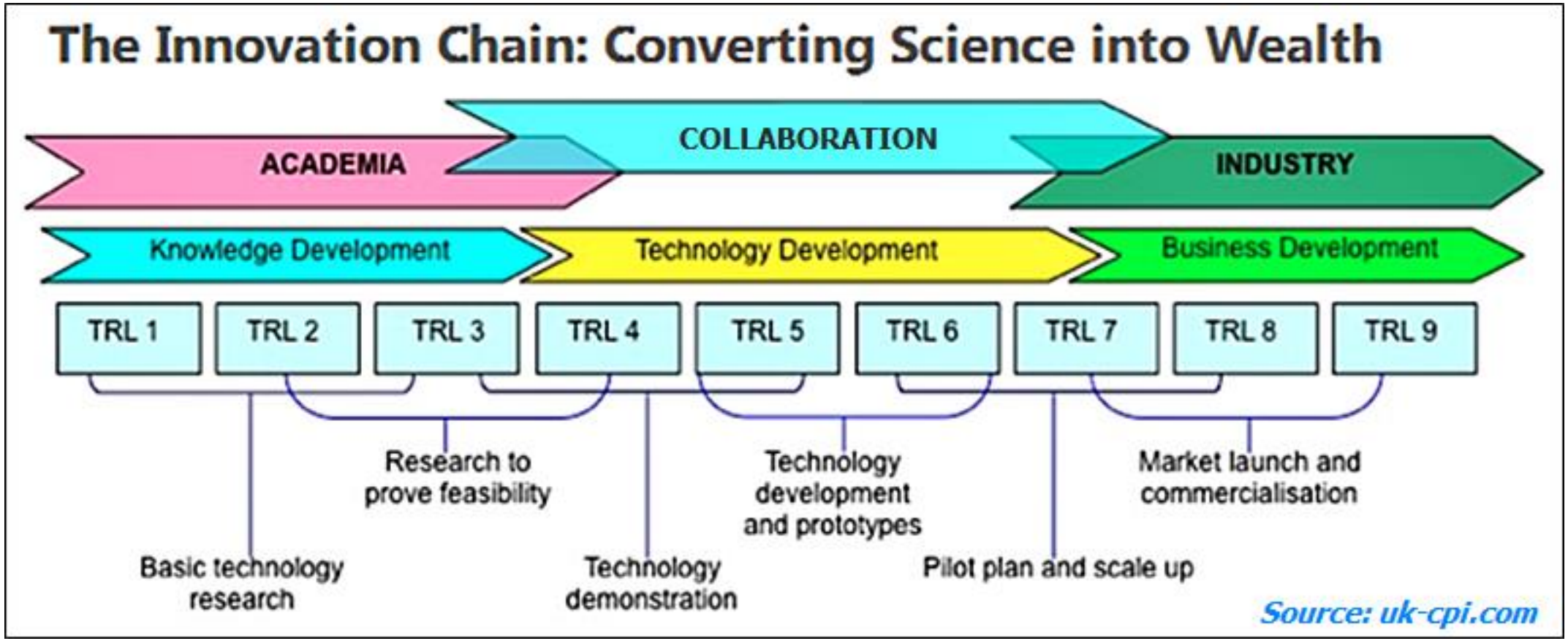
«real» problems to research!

Research with industrial partners solves real problems

and

naturally leads to licensing.

Technology Readines Level -TRL



Types of Relationships with Industry

- *Pre-competitive research*
- ***Contracted Research***
- ***Collaborative Research***
- ***Knowledge transfer partnerships***

A New Model for Pre-Competitve Partnerships

Phase 1:
Collaborative Networks

Interface Supports

OUTPUTS:
Product/Theme Roadmaps
Strategic Partnerships

Phase 2:
Strategic Products

Mega R&D Supports

OUTPUTS:
Strategic Products/Strategic Technologies
Sharing Knowledge/Experiences

HUB1 Lead By Research Institutes

Phase 3:
Industrial Innovation Partnerships

R&D Supports for
Networks in Priority Areas

OUTPUTS:
THS 6-9
Strategic Products
in Priority Areas

HUB2 Lead By Industrial Companies

Fundamental Research Feeding "HUB 1"

TRL 4-6 Level Research from HUB 1 Feeding "HUB 2"



Technology Readines Level -TRL

